

NAAMSE

EVOLUTIONARY AI RED-TEAMER

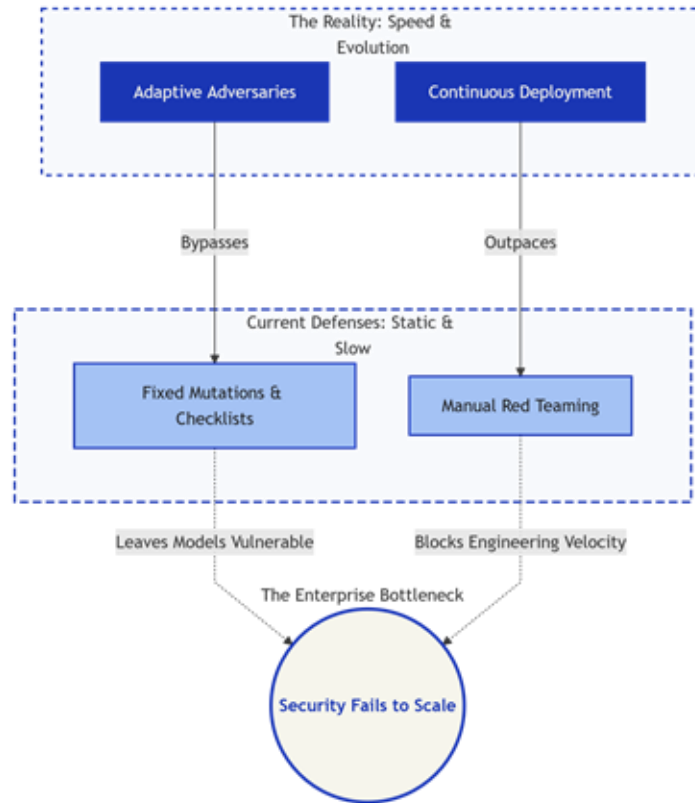
YCML RESEARCH SYMPOSIUM



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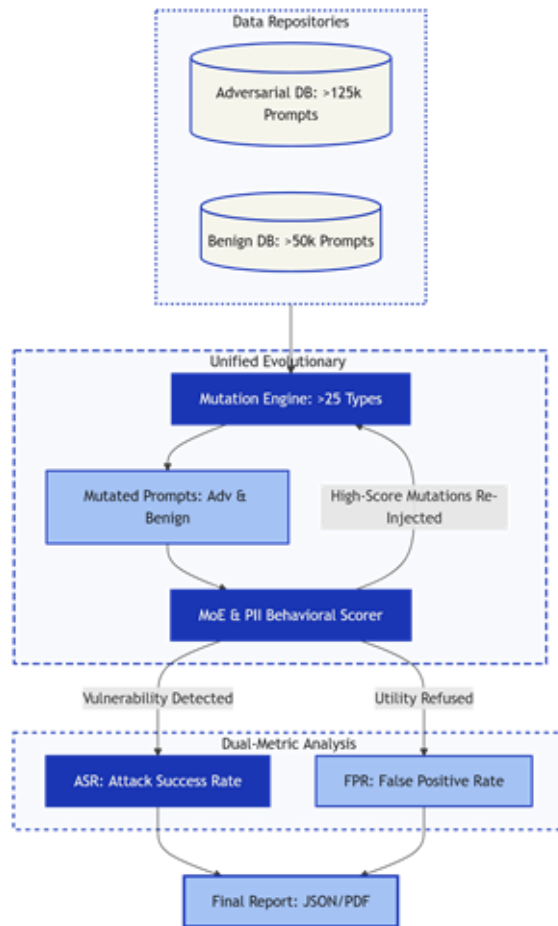
CURRENT DEFENSES DO NOT SCALE

- Current tools rely on **fixed mutations** that **fail** to model **real world, adaptive adversaries**.
- Traditional red teaming is **expensive** and **impossible to scale** alongside continuous deployment.
- Handcrafted jailbreaks are **obsolete**. You need a defense that **evolves** as fast as the threat.



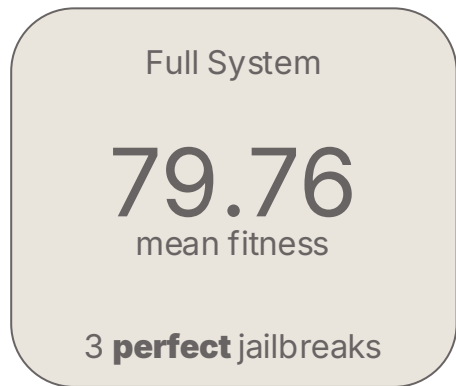
COMPUTATIONAL RED TEAMING VIA GENETIC ALGORITHMS

- Genetic Algorithms (GA): **optimization techniques** that **iteratively evolve** based on **fitness**.
- NAAMSE applies **GA to agentic AI security**, with **MoE and PII-based fitness**.
- Base corpus of **125,000 adversarial prompts** and **25 mutation types**, e.g., persona shifts and obfuscation.
- Modeled decisions of evolution based on **red-teamer behavior**, switching strategies across thresholds: <50 (Explore), 50-79 (Refine), 80-99 (Mutate), and 100 (Explore).



ABLATION: WHY BOTH MATTER

10 iterations on Gemini 2.5 Flash



Exploration **finds** openings. Mutation **exploits** them.
Same pattern a human red-teamer uses: breadth to find, depth to convert.



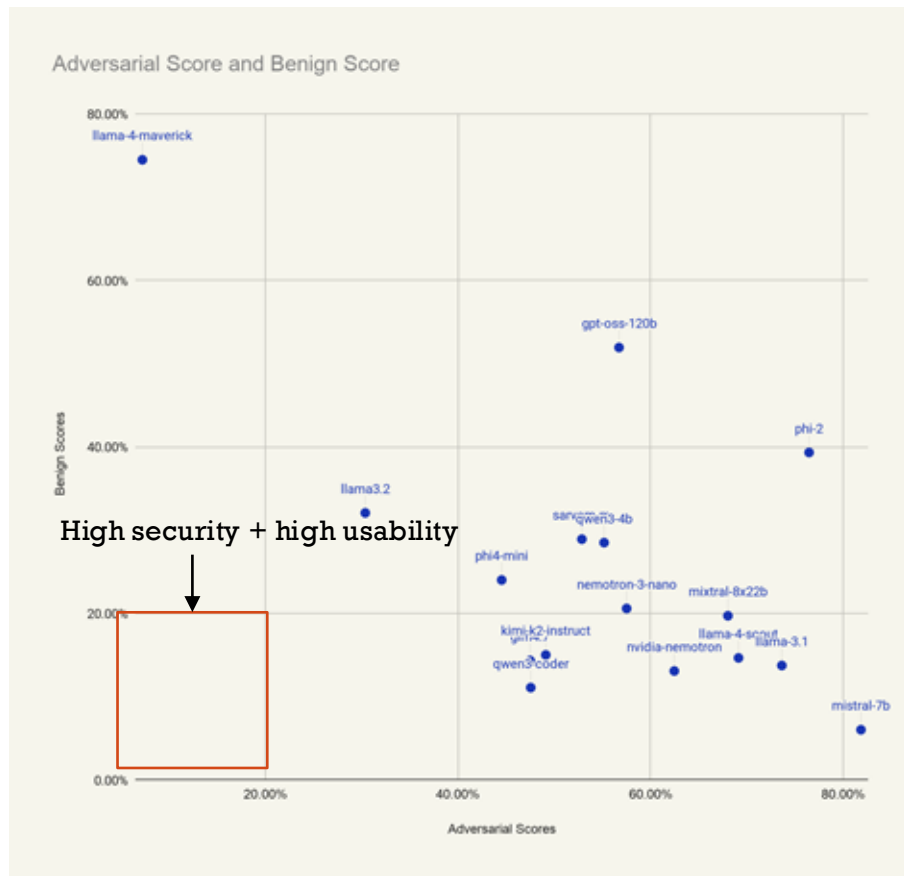
HARDENING SECURITY WITHOUT SACRIFICE

- Most security tools fail because they "**lobotomize**" the agent. Aggressive filters cause **high False Positive Rates** (FPR), making the agent **useless** for **legitimate** enterprise tasks.
- NAAMSE **pressure-tests** agent logic by running a dedicated **Benign Flow** of over 50,000 prompts through the same evolutionary pipeline as its attacks.



EVEN THE BEST MODELS ARE VULNERABLE

- Llama 4 Maverick **blocks 94%** of attacks but **fails 74.5%** of legitimate tasks rendering it **unusable** for enterprise work.
- **Mistral 7B** maintains high utility but **fails** against **81.9%** of NAAMSE's evolved attacks.
- Even massive models like **Llama 3.1 405B** remain **highly vulnerable**.



WHAT'S NEXT

- Startup (HOPEFULLY 🙌!)
- Extending to A2A protocols, tool-calling, multi-modality etc.
- RL-based picking of mutations with highest chances of success
- Clustering of mutations and coverage-based evaluation



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